

WONG, Hok Fong

Intern, Summer@EPFL; BSc in Computer Science, CUHK

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ACADEMIC INTERESTS

Machine Learning and Theory; Optimization; Design and Analysis of Algorithms; Theoretical Computer Science; Atmospheric Science

EDUCATION

The Chinese University of Hong Kong, Hong Kong

[Academic Transcript]

Aug 2022 - Jun 2026

Major: **B.Sc. in Computer Science** with **First-Class Honors** (Specialization: Algorithms and Complexity)

Cumulative GPA: **3.803 / 4.0**

Minors: **Mathematics; Earth and Environmental Sciences** (Focus: Atmospheric Science)

151 Units = ~302 ECTS [Ref]

Engineering Enhancement Program: **Engineering Leadership, Innovation, Technology and Entrepreneurship Stream (ELITE Stream)**

PUBLICATION

H. F. Wong, H.-T. Wai, C.-Y. Yau, "Large Stepsizes Federated Learning on Logistic Regression with Linearly Separable Data: The Case of Heterogeneous Devices," *The 65th IEEE Conference on Decision and Control* ([accepted](#)), 2026.

SELECTED ACADEMIC AWARDS

Best Project Award - UG Summer Research Internship 2025 (Faculty of Engineering, CUHK)

2025

- Ranked top 10% among all undergraduate summer research projects, based on originality, analytical rigor, and professional delivery of the project "Revisiting ASkewSGD: New Theoretical Guarantees for Quantization-Aware Deep Neural Network Optimization".
- Faculty Website (Web Archive): [Link](#)

Andrew C. Yao Fellowship – HKD \$50,000

2024

- Awarded to only 2 outstanding CSE undergraduates annually based on academic distinction and exceptional extracurricular achievements.
- Highest distinction in CSE department, named in honor of Professor Andrew Chi-Chih Yao (Turing Award, 2000).
- Department Scholarship Website (Hall of Fame): [Link](#)

ELITE Stream Scholarship – HKD \$20,000

2024, 2026

- Awarded to students of the Engineering Leadership, Innovation, Technology and Entrepreneurship Stream with outstanding performance in recognized engineering-discipline courses with challenging coursework.
- Scholarship Description: [Link](#)

CUHK Vice-Chancellor's Scholarships for Excellence – HKD \$50,000

2023

- One of the only 8 first-year undergraduate students selected university-wide for exceptional academic and distinguished extracurricular achievements, recognizing top-tier students with outstanding potential.
- Official Press Release: [Link](#)

HKSAR Government Talent Development Scholarship – HKD \$10,000

2023

- For the exceptional achievements in the innovation, science and technology area (non-academic).
- Official Information: [Link](#)

Dr. Stanley Ho Medical Development Foundation Scholarship – HKD \$18,000 (renewable)

2022, 2023, 2024, 2025

- Awarded to CUHK students from Macao with excellent academic performance (offered by Morningside College, CUHK).
- College News: [Link](#)

Dean's List (Faculty of Engineering, CUHK)

2023, 2024, 2025, 2026

- Awarded to Engineering students with excellent academic performance (Top 10%).

SELECTED PROJECTS

Theoretical Analysis of Large-Stepsize Gradient Descent on Multinomial Logistic Regression

Fall 2025, Spring 2026

- Graduation Thesis challenging conventional smoothness-based convergence limits for large-stepsize Gradient Descent. [PDF](#)
- Pioneered a novel theoretical analysis of the optimization dynamics of multinomial logistic regression, delivering its first non-asymptotic convergence bound. Precisely quantified that both loss and phase transition time scale only logarithmically with the number of classes.
- Extended analysis to FedAvg with large stepsizes and heterogeneous local updates where heterogeneity vanished asymptotically. Also characterized convergence on non-separable data: adaptive stepsizes guarantee monotonic $O(1/t)$ convergence, constant-stepsize cycles exist for $K \geq 3$ classes.

On Quantization-Aware Training Algorithms and Stronger Convergence Guarantees of ASkewSGD

Summer 2024, 2025

- Research project focused on improving theoretical guarantees of the ASkewSGD algorithm for quantizing deep neural networks (DNNs).
- Eliminated the need for convexity, Lipschitz continuity and differentiability requirements commonly adopted in prior work via Stochastic Approximation.
- Demonstrated ASkewSGD's superiority over benchmarks (BinaryConnect, ProxQuant) in tasks such as synthetic data (TwoMoons) and image classification (MNIST, CIFAR-10) with low-bit quantization for resource-efficient DNN deployment. We publicized reproducible experiments on [GitHub](#).

Frequency Estimation in Strict-Turnstile Data Streams with L_1 α -Bounded Deletions

Spring 2026

- Extended four classical streaming algorithms (Misra-Gries, SpaceSaving, Count-Min Sketch, Count-Sketch) to the strict turnstile model with the L_1 α -bounded deletion property, proving tighter error bounds that scale with α instead of stream length.
- Conducted extensive experiments on Zipfian, uniform, and real-world CAIDA network traces (sliding window and idle timeout streams).
- Proposed a novel algorithm, Dyadic Count-Min Hierarchy with Adaptive Counter Cache, a hybrid that combines logarithmic space sampling with a small deterministic cache, preserving counter-based accuracy for the heaviest items while remaining space efficient for large α .
- Project available on [GitHub](#)

First-Order Methods in Online Learning and Online Convex Optimization

Fall 2024

- Studied and implemented key algorithms such as Online Gradient Descent (OGD), Follow-the-Regularized-Leader (FTRL), and Online Mirror Descent (OMD), analyzing their regret bounds and convergence behavior under adversarial settings with strong convexity and smoothness assumptions.
- Conducted empirical evaluations comparing algorithmic performance across synthetic and real-world datasets, focusing on regret minimization and computational efficiency for online learning / online convex optimization. The project is available on [GitHub](#).

Field Journal: Civic Engagement and Humanitarian Actions for Refugees in Athens and Lesbos

Fall 2024

- Authored a comprehensive, critical journal analyzing the operational efficacy, funding models, and legal challenges of 5+ humanitarian NGOs (e.g., Za'atar, KHORA, Doctors of the World) in Greece, based on direct field engagement and qualitative observation.
- Engaged in advanced reflective practice on the ethics of international service-learning, critically examining power dynamics, the potential harms of "voluntourism," and the responsibility of global citizenship. Synthesized interdisciplinary learning from history, law, and political philosophy to contextualize contemporary migration issues, drawing connections between ancient Greek "metics" and modern asylum seekers' legal status. [PDF version](#)

A Practical Test Set Bound in Binary Classification Theory

Fall 2023

- Reconstructed and analyzed key proofs by John Langford to deepen understanding of generalization bounds for binary classification based on merely the empirical risk and size of the training set, grounded in principles from statistical learning theory.
- Implemented simulations to apply the proposed bounds under given conditions of proposed model performance. The project is available on [GitHub](#).

A detailed list of projects, essays, and codes are available --- [Link](#)

WORK EXPERIENCES

(Ongoing) Summer Research Fellow – Summer@EPFL 2026, Switzerland

Jun 2026 – Aug 2026

- Selected from 7,000+ global applicants (**1.2% acceptance rate**), with the focus on **optimization for machine learning (theory-focused)**.

Research Student – CUHK Undergraduate Summer Research Internship (2024, 2025)

Jun 2024 – May 2026

- Conducted research in modern optimization methods tailored for **large-scale machine learning**, focusing on stochastic methods for nonconvex problems.
- Analyzed theoretical properties such as **convergence guarantees, and complexity bounds** under smoothness and/or bounded gradient assumptions.
- Awarded **research grants** from Faculty of Engineering, with a total amount of **HKD \$28,000**.
- Supervisor: Prof. Hoi-To WAI, Department of System Engineering and Engineering Management, CUHK

Curriculum Designer – CUHK & Jockey Club “AI for the Future” Project

Mar 2023 – Dec 2024

- Spearheaded the **creation of engaging courseware** introducing the knowledge of Artificial Intelligence (AI) to thousands of secondary school students.
- Collaborated with professors and post-doctoral researchers to design curriculum content aligned with the **first AI syllabus launched for junior secondary education in Hong Kong**. [\[Website\]](#)
- Received stipends with a total amount of **HKD \$12,000** for the technical quality and the education impact of the work conducted.
- Supervisor: Prof. Mei Ling Helen MENG, Dr. Xing XING, Stanley Ho Big Data Decision Analytics Research Centre, CUHK

COMPETITIONS

Huawei ICT Competition (Hong Kong Regional, Final Round) - Cloud Track

Nov 2022 – Dec 2022

- **Potential Award** with **HKD \$1,500**, for knowledge and skills related to cloud computing and Huawei's cloud technologies.

International Collegiate Programming Contest (ICPC), Asia Macau Region

May 2021, Apr 2022

- Awarded **Outstanding High School Team** with **MOP \$1,000** (for both years).

Macao Olympiad in Informatics

2018, 2019, 2020, 2021, 2022

- Consistent remarkable performance with an accumulation of **4 Gold Medals (Top 5%)** and **1 Silver Medal** from the first participation onward. [\[Link\]](#)
- Designed, implemented, and tested **efficient algorithms and data structures** in C++ to solve problems under time and memory constraints with **strong logical reasoning and programming proficiency**.
- Competed in **National Olympiad in Informatics (CCF NOI '19)**, **International Olympiad in Informatics (IOI '21, '22)**, **Team of Macao**. [\[Link\]](#)

LEADERSHIP & EXTRACURRICULARS

Vice Chairman and Secretary General – The Macao Association for Competitive Programming

Apr 2026 – Apr 2029 (Est.)

- Government recognized [registered association](#), with the goal of cultivating students' algorithmic thinking through seminars, invited talks, and contests.
- Co-organized the “HKACE 45th Anniversary Cup – HKOI 2026 Asia Invitational” in the Macao Region with Macao Computer Society in May 2026.
- Official Website: [\[Link\]](#)

Participant – Greece Service-Learning Trip (Organized by Morningside College, CUHK)

June 2024

- Engaged with non-governmental organizations ([Doctors of the World](#), etc.) with the focus on the **Greek NHS and healthcare access for refugees**.
- Performed **critical evaluations** related to **operation structures of NGOs** and how **challenges related to refugees and asylum seekers** are tackled.
- Authored a **comprehensive 10,000-word field journal** and submitted a formal **evaluation report** proposing actionable improvements to healthcare delivery and NGO coordination. Related documents are available [here](#) and [here](#).
- Reference: Morningside College, CUHK ([College News Highlight](#))



Secretary – The 12th Morningside College Student Union (Executive Council)

Mar 2023 – Feb 2024

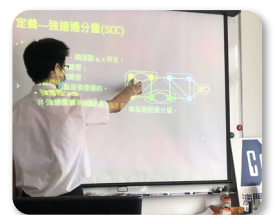
- Maintained accurate meeting minutes and digitalized organizational records, ensuring transparency and continuity across student-led operations.
- Excelled in **drafting activity proposals and managed complex logistics**.
- Led initiatives that fostered **inclusivity and cross-cultural engagement** that enhanced the well-being of Morningside College's diverse and international student community.
- **Organized events and activities** including Graduation Photo Day, Mid-Autumn Festival, Halloween Party, Cultural Fair, Poon-choi Dinner **under strict budget constraints**.
- Reference: Morningside College, CUHK ([Event 1](#) [Event 2](#) [Event 3&4](#) [Event 5](#))



Member – Macao Computer Society

Sep 2020 – Present

- **Provided comprehensive training programs** online and offline for the Macao team at the International Olympiad in Informatics (IOI) at year 2022, 2023. [\[Selected Session's Slides\]](#)
- **Designed the banner and served as an invigilator for Macao Olympiad in Informatics (MOI) in 2023**, as a member of the organizing committee. [\[Banner\]](#) [\[Ceremony\]](#)
- Served as **invigilator for Macao Olympiad in Artificial Intelligence (MOAI) in 2026**.
- **Proposed 2 original and challenging ICPC-styled competitive programming problems to the Joint-School Invitational Contest** in 2022, designed to test algorithmic thinking and problem-solving skills among secondary school participants, fostering a sense of community and healthy competition.
 - **Problem 1 (Prime Maze)**: *Observations in number theory and search algorithms*, requiring solvers to identify mathematical patterns to avoid inefficient brute-force solutions. [\[Editorial Available\]](#)
 - **Problem 2 (Matchmaker)**: *Basic number theory with (advanced) data structure implementations*.
- Reference: Mr. Kuo-Meng SUN (Vice-executive director of MCS),



TECHNICAL SKILLS

Language: Cantonese [Native], Mandarin [Native], English [Proficient] (IELTS Academic 7.5, L-9.0 R-8.5 W-6.5 S-6.5, Test date: Jun 2024).

Programming: C, C++, Java, Python{PyTorch, pandas, openpyxl, scikit-learn, Matplotlib, Django}, OCaml, Prolog, JavaScript, HTML, CSS.

Tools: SQL, LaTeX, Git, WandB, Linux, GEOS-Chem.

IT Certifications: LPI Linux Essentials 010-160 (Score: 800/800); CCF Assessment of Programming Proficiency Level for Secondary Students – Level 7.

Competitive Programming: Efficient and Optimized Computations, Algorithmic Thinking & Problem-Solving, Pedagogical Skills.

COURSEWORKS

Mathematics: Single/Multi-variable Calculus, Linear Algebra, Discrete Mathematics, Probability and Statistics, Game Theory, Ordinary Differential Equations, Optimization Theory, Mathematics for Data Analytics.

Computer Theory: Data Structures and Algorithms, Formal Languages and Automata Theory, Principles of Programming Languages, Computational Geometry (Postgraduate-Level), Algorithms for Big Data (Postgraduate-Level).

AI&ML: Fundamentals of Artificial Intelligence, Fundamentals of Machine Learning, Numerical Optimization.

Systems: Digital Logic and Systems, Computer Organization, Software Engineering, Operating Systems.

Earth Science: Climate System Dynamics, Solid Earth Dynamics, Ecological Climatology, Cloud Dynamics, Atmospheric Chemistry and Modeling.

Last updated: Jul 18, 2026